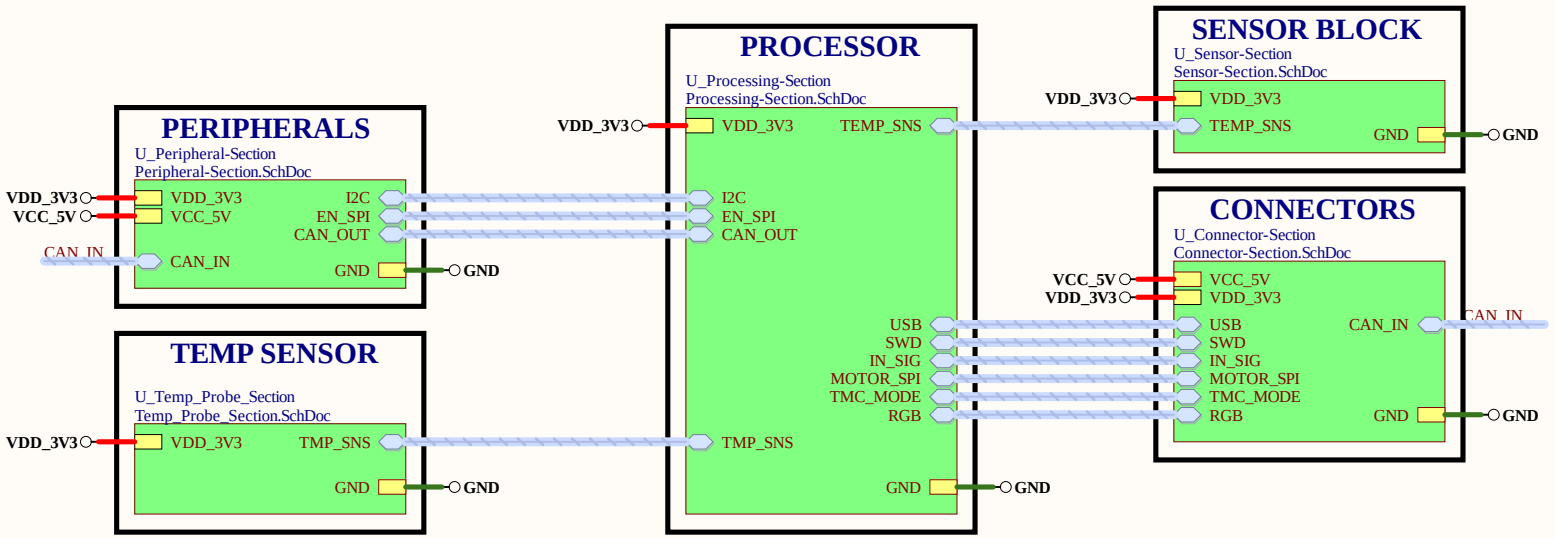
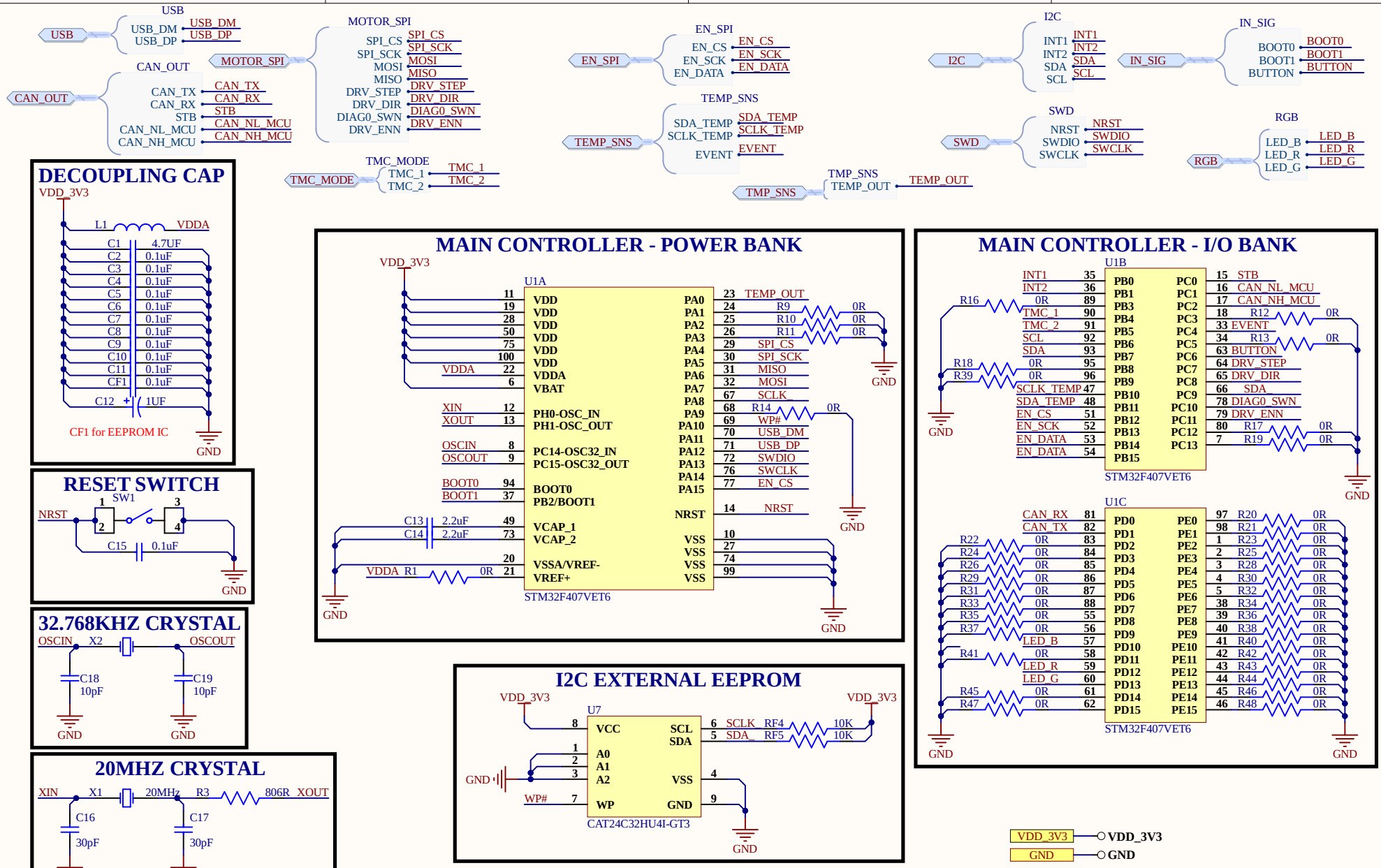
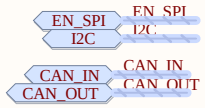
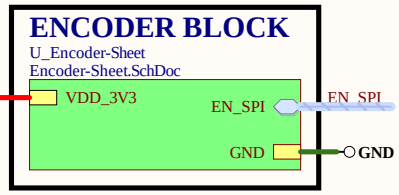
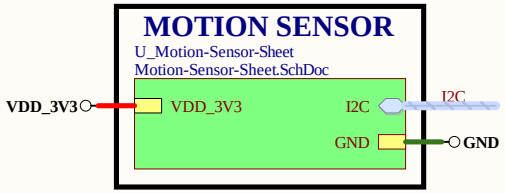
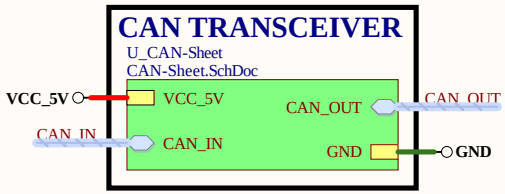

PERIPHERALS
U_Peripheral-Section
Peripheral-Section.SchDoc
VDD_3V3
VCC_5V
CAN_IN
VDD_3V3
VCC_5V
CAN_IN
I2C
EN_SPI
CAN_OUT
GND
GND
TEMP SENSOR
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
VDD_3V3
TMP_SNS
GND
GND
PROCESSOR
U_Processing-Section
Processing-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
I2C
EN_SPI
CAN_OUT
TMP_SNS
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
GND
GND
SENSOR BLOCK
U_Sensor-Section
Sensor-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VCC_5V
VDD_3V3
VCC_5V
VDD_3V3
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
CAN_IN
GND
GND
CAN_IN

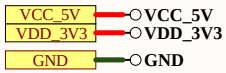

USB
USB_DM
USB_DP
CAN_OUT
CAN_TX
CAN_RX
STB
CAN_NL_MCU
CAN_NH_MCU
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TMC_MODE
TMC_1
TMC_2
EN_SPI
EN_CS
EN_SCK
EN_DATA
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
TMP_SNS
TEMP_OUT
I2C
INT1
INT2
SDA
SCL
SWD
NRST
SWDIO
SWCLK
IN_SIG
BOOT0
BOOT1
BUTTON
RGB
LED_B
LED_R
LED_G
DECOUPLING CAP
VDD_3V3
L1
VDDA
C1 4.7UF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
CF1 0.1uF
C12 1UF
CF1 for EEPROM IC
GND
RESET SWITCH
SW1
NRST
C15 0.1uF
32.768KHZ CRYSTAL
OSCIN
X2
OSCOUT
C18 10pF
C19 10pF
20MHZ CRYSTAL
XIN
X1
20MHz
R3 806R
XOUT
C16 30pF
C17 30pF
MAIN CONTROLLER - POWER BANK
U1A
VDD
VDDA
VBAT
PH0-OSC_IN
PH1-OSC_OUT
PC14-OSC32_IN
PC15-OSC32_OUT
BOOT0
PB2/BOOT1
VCAP_1
VCAP_2
VSSA/VREF-
VREF+
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
NRST
VSS
TEMP_OUT
SPI_CS
SPI_SCK
MISO
MOSI
SCLK
WP#
USB_DM
USB_DP
SWDIO
SWCLK
EN_CS
R9 0R
R10 0R
R11 0R
R14 0R
C13 2.2uF
C14 2.2uF
VDDA_R1 0R
STM32F407VET6
MAIN CONTROLLER - I/O BANK
U1B
U1C
STB
CAN_NL_MCU
CAN_NH_MCU
R12 0R
R13 0R
DRV_STEP
DRV_DIR
SDA_
DIAG0_SWN
DRV_ENN
R17 0R
R19 0R
R16 0R
R18 0R
R39 0R
R20 0R
R21 0R
R22 0R
R23 0R
R24 0R
R25 0R
R26 0R
R28 0R
R29 0R
R30 0R
R31 0R
R32 0R
R33 0R
R34 0R
R35 0R
R36 0R
R37 0R
R38 0R
R40 0R
R41 0R
R42 0R
R43 0R
R44 0R
R45 0R
R46 0R
R47 0R
R48 0R
I2C EXTERNAL EEPROM
U7
VCC
A0
A1
A2
WP
SCL
SDA
VSS
GND
SCLK_RF4 10K
SDA_RF5 10K
CAT24C32HU4I-GT3
VDD_3V3
GND


EN_SPI
EN_SPI
I2C
I2C
CAN_IN
CAN_IN
CAN_OUT
CAN_OUT

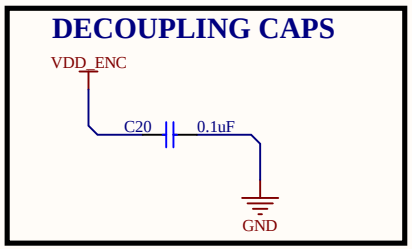

ENCODER BLOCK
U_Encoder-Sheet
Encoder-Sheet.SchDoc
VDD_3V3
VDD_3V3
VDD_3V3
EN_SPI
EN_SPI
GND
GND
GND

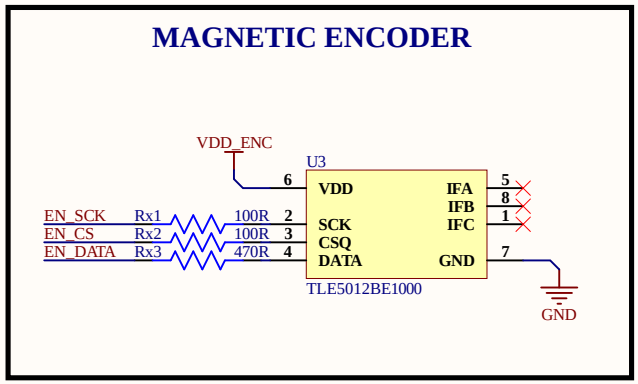

MOTION SENSOR
U_Motion-Sensor-Sheet
Motion-Sensor-Sheet.SchDoc
VDD_3V3
VDD_3V3
VDD_3V3
I2C
I2C
GND
GND
GND

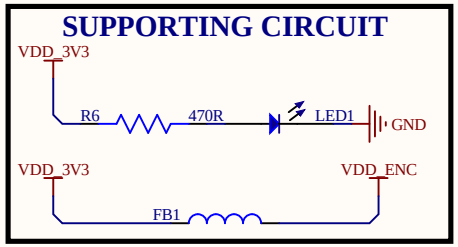

CAN TRANSCEIVER
U_CAN-Sheet
CAN-Sheet.SchDoc
VCC_5V
VCC_5V
VCC_5V
CAN_OUT
CAN_OUT
CAN_IN
CAN_IN
CAN_IN
GND
GND
GND

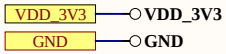

VCC_5V
VCC_5V
VDD_3V3
VDD_3V3
GND
GND

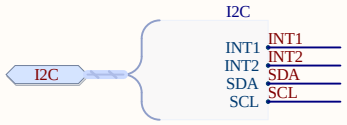

EN_SPI
EN_SPI
EN_CS
EN_SCK
EN_DATA
EN_CS
EN_SCK
EN_DATA


DECOUPLING CAPS
VDD_ENC
C20
0.1uF
GND

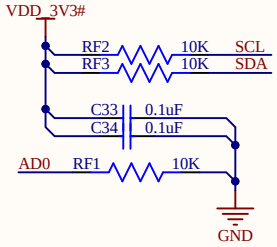

MAGNETIC ENCODER
VDD_ENC
U3
6
VDD
IFA
IFB
IFC
5
8
1
EN_SCK
Rx1
100R
2
SCK
EN_CS
Rx2
100R
3
CSQ
EN_DATA
Rx3
470R
4
DATA
GND
7
TLE5012BE1000
GND


SUPPORTING CIRCUIT
VDD_3V3
R6
470R
LED1
GND
VDD_3V3
VDD_ENC
FB1


VDD_3V3
VDD_3V3
GND
GND



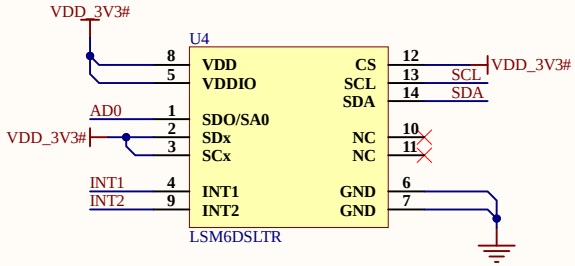
SUPPORTING CIRCUIT

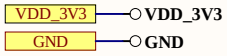


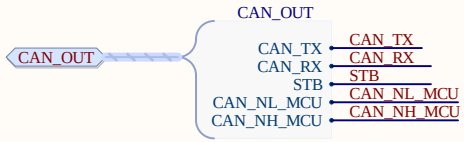
FERRITE CONNECTION

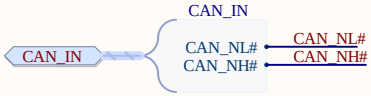


6-AXIS MOTION SENSOR

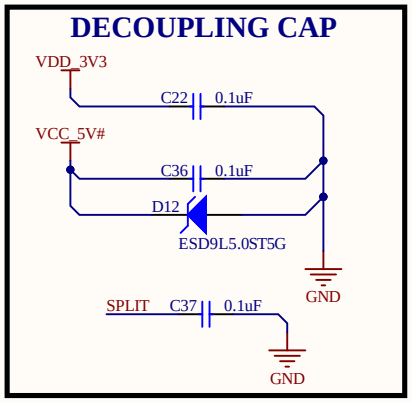


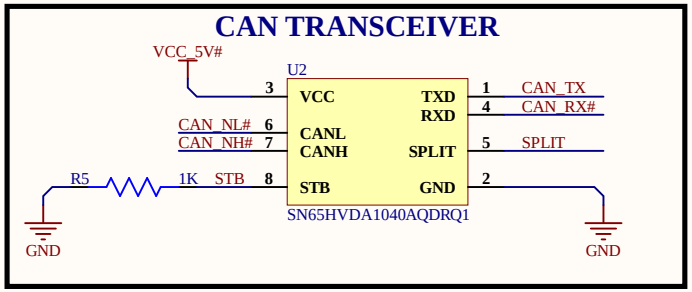


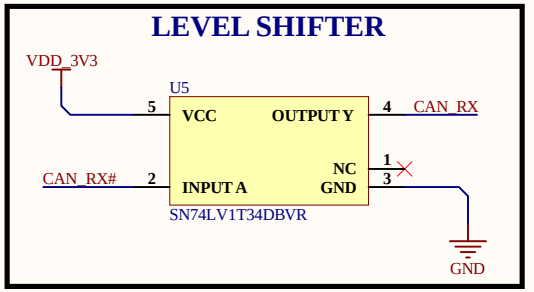


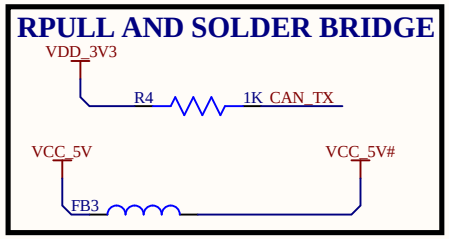


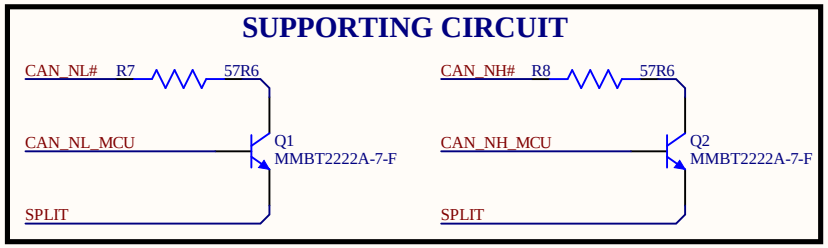
Connections for CAN are w.r.t Controller Rx and Tx

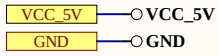


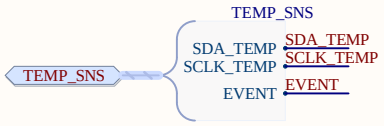


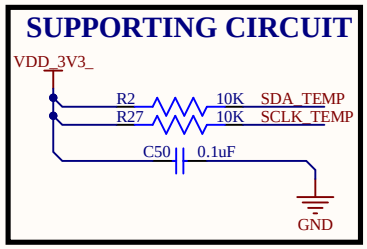


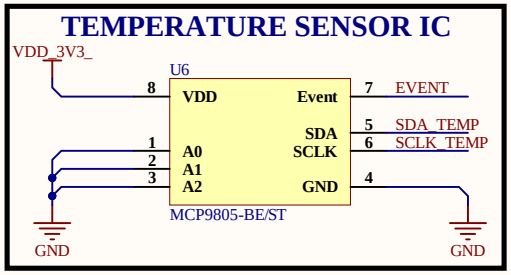




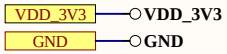



TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SDA_TEMP
SCLK_TEMP
EVENT


SUPPORTING CIRCUIT
VDD_3V3_
R2
10K
SDA_TEMP
R27
10K
SCLK_TEMP
C50
0.1uF
GND

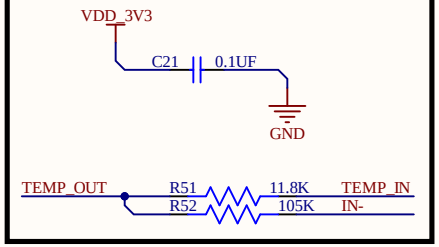

TEMPERATURE SENSOR IC
VDD_3V3_
U6
8
VDD
Event
7
EVENT
SDA
5
SDA_TEMP
1
A0
SCLK
6
SCLK_TEMP
2
A1
3
A2
GND
4
MCP9805-BE/ST
GND
GND


FERRITE CONNECTION
VDD_3V3
VDD_3V3_
FB4

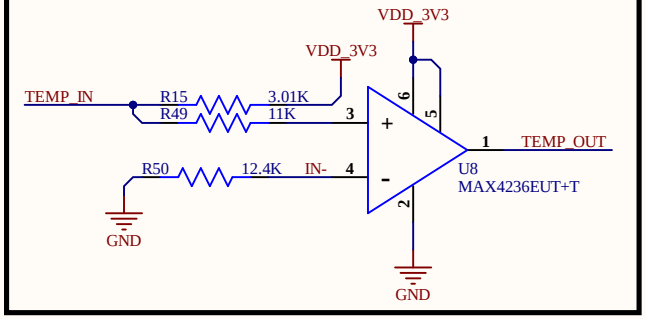

VDD_3V3
VDD_3V3
GND
GND



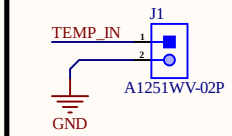
SUPPORTING CIRCUIT

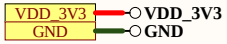


TEMPERATURE SENSOR

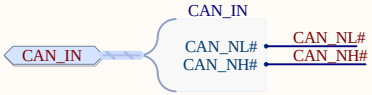


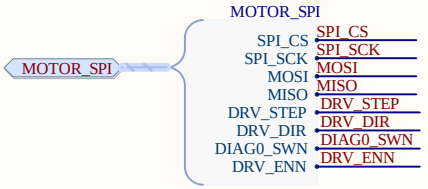
PT100 INPUT



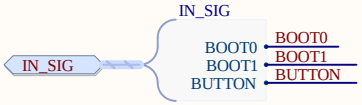


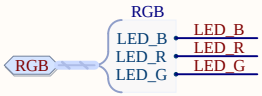

USB
USB_DM
USB_DP
USB_DM
USB_DP
USB


CAN_IN
CAN_NL#
CAN_NH#
CAN_NL#
CAN_NH#
CAN_IN

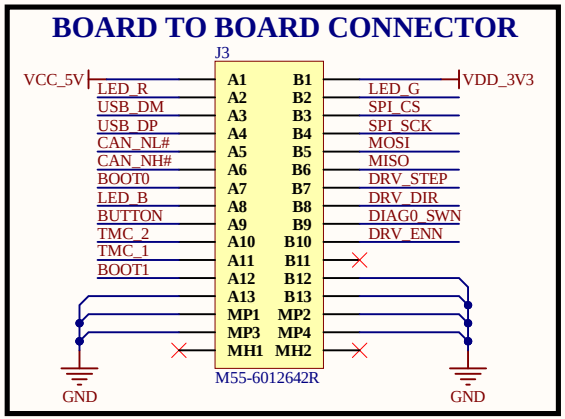

MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
MOTOR_SPI

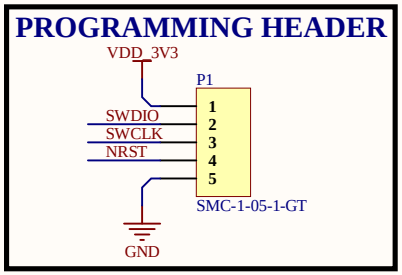

TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2
TMC_MODE

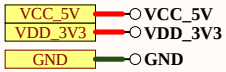

IN_SIG
BOOT0
BOOT1
BUTTON
BOOT0
BOOT1
BUTTON
IN_SIG

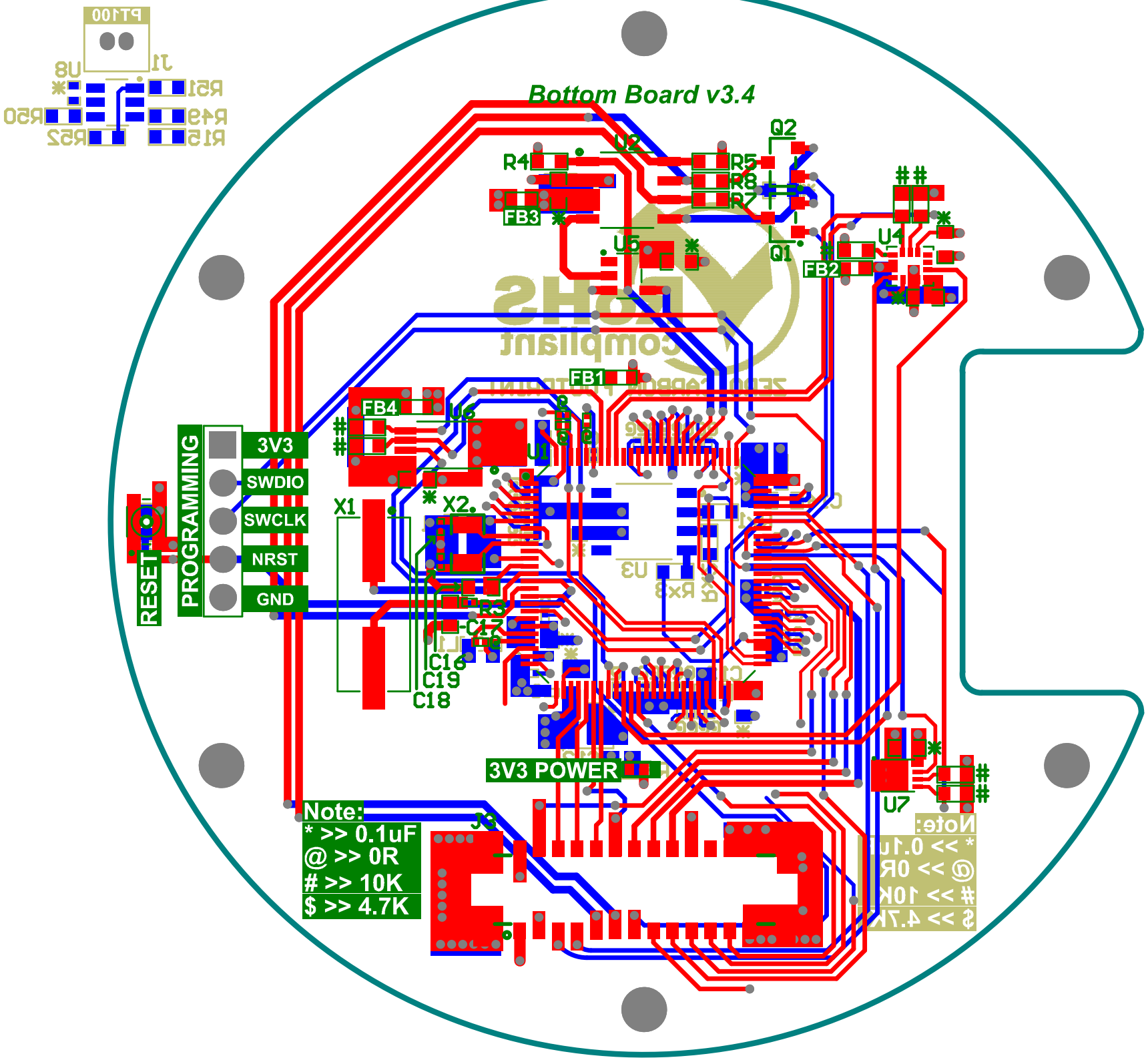

RGB
LED_B
LED_R
LED_G
LED_B
LED_R
LED_G
RGB


SWD
NRST
SWDIO
SWCLK
NRST
SWDIO
SWCLK
SWD


BOARD TO BOARD CONNECTOR
J3
VCC_5V
VDD_3V3
LED_R
LED_G
USB_DM
SPI_CS
USB_DP
SPI_SCK
CAN_NL#
MOSI
CAN_NH#
MISO
BOOT0
DRV_STEP
LED_B
DRV_DIR
BUTTON
DIAG0_SWN
TMC_2
DRV_ENN
TMC_1
BOOT1
A1 B1
A2 B2
A3 B3
A4 B4
A5 B5
A6 B6
A7 B7
A8 B8
A9 B9
A10 B10
A11 B11
A12 B12
A13 B13
MP1 MP2
MP3 MP4
MH1 MH2
M55-6012642R
GND
GND


PROGRAMMING HEADER
VDD_3V3
P1
SWDIO
SWCLK
NRST
1
2
3
4
5
SMC-1-05-1-GT
GND


VCC_5V
VDD_3V3
GND
VCC_5V
VDD_3V3
GND


Bottom Board v3.4
PROGRAMMING
3V3
SWDIO
SWCLK
NRST
GND
RESET
3V3 POWER
Note:
* >> 0.1uF
@ >> 0R
>> 10K
$ >> 4.7K
Q2
Q1
U2
U4
U5
U6
U7
U1
R4
R5
R8
R7
FB1
FB2
FB3
FB4
X1
X2
C16
C19
C18
R3
J3